

Olympic Sports Performance Dashboard Report

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Project Title: Olympic Sports Performance Dashboard

Dataset: 120 Years of Olympic History (1896–2016), 217117 rows and 18 columns

Data Source: Kaggle – Olympic History Dataset

https://www.kaggle.com/datasets/mysarahmadbhat/120-years-of-olympic-history?select=athlete_events.csv

Business Problem

National Olympic Committees and sport's governing bodies require a deeper understanding of the factors that drive Olympic success. However, the large volume of historical Olympic data makes it difficult to identify trends in athlete participation, medal performance, country rankings, and sport specialization. This project develops an interactive Power BI dashboard that analyses Olympic data from 1896 to 2016 to uncover patterns in athlete characteristics, participation trends, medal distribution, and country performance. The insights generated can support strategic decisions related to athlete development, resource allocation, and long-term sports performance planning.

Business Questions

The dashboard was developed to answer the following questions:

1. Which countries have won the most Olympic medals?
2. Which sports generate the highest medal counts?
3. How has Olympic participation changed over time?
4. How does male and female participation compare across years?
5. Which countries consistently perform well in specific sports?
6. What is the trend of medal performance over time?
7. Which age groups achieve the highest Olympic success?
8. How do Summer and Winter Olympics differ in participation and medal distribution?

Dashboard Overview

The dashboard consists of four main pages:

- Executive Summary
- Trend Analysis
- Athlete Profile Analysis
- Country & Sport Performance Analysis

An additional Athlete Details Drill-through Page was created to allow users to explore individual athlete information.

Key Insights

Insight 1: Country Medal Dominance

The United States is the most successful Olympic nation with approximately 5.6K medals, significantly outperforming other countries such as URS, Germany, Great Britain, and France.

Insight 2: Top Medal-Winning Sports

Athletics recorded the highest medal count, followed by Swimming and Rowing. These sports contribute a substantial share of total Olympic medals.

Insight 3: Growth of the Olympic Games

Athlete participation and medal counts have increased significantly over time, indicating the continued growth and global expansion of the Olympic Games.

Insight 4: Peak Athlete Age

Athletes aged 20–29 years account for most Olympic participants, while the average athlete age is approximately 24.45 years, suggesting that peak Olympic performance occurs during this age range.

Insight 5: Concentration of Olympic Success

Olympic success is concentrated among a relatively small number of countries. Nations such as the United States, France, Great Britain, Germany, and Italy consistently achieve strong medal performances.

Recommendations

Recommendation 1

Increase investment in high-performing sports such as Athletics, Swimming, and Rowing, as these sports provide the greatest opportunities for medal success.

Recommendation 2

Strengthen youth talent identification and athlete development programmes to ensure athletes reach peak performance during the 20–29 age range.

Recommendation 3

Benchmark against consistently successful countries such as the United States and Great Britain to identify effective athlete development strategies.

Recommendation 4

Use athlete characteristics such as age, height, weight, and BMI to support sport-specific talent identification and training programmes.

Recommendation 5

Adopt data-driven monitoring tools such as Power BI dashboards to continuously track athlete participation, performance trends, and medal outcomes.

Conclusion

The Olympic Sports Performance Dashboard provides valuable insights into historical Olympic participation, athlete characteristics, medal distribution, and country performance. The analysis shows that Olympic success is concentrated among a few countries, Athletics remains the most successful sport, and athletes aged 20–29 dominate Olympic participation. These findings can support evidence-based decision-making for sports organizations, coaches, and policymakers seeking to improve future Olympic performance.